The Factor Statistics of Twin Centres on the $6N$ Skeleton: Each Prime Divides with Probability $1/(q-2)$, not $1/q$

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Abstract

Throughout Parts I–XI the factor count $\omega_{>3}(N)$ was used as a given stratum. Here we ask what distribution it follows for the sparse set of twin centres themselves. The answer is a single clean replacement: a prime $q > 3$ divides a twin centre with probability $1/(q-2)$, against $1/q$ for an ordinary integer. We verify this per prime on the 2.4×10^7 twin centres of S_{10} (and S_9): for small q the measured $P(q \mid N,\, N \text{ twin})$ equals $1/(q-2)$ to within 0.1%. This is the microscopic root of the Part I enrichment factor $\prod(q-1)/(q-3)$, which is exactly the product of these raised single-prime probabilities. The consequent rightward shift of the mean has the convergent closed form $\sum_{q>3}\left(\frac{1}{q-2} - \frac{1}{q}\right) = 0.2604$; the finite-$N$ measured shift increases with the shell ($+0.222$ on S_9, $+0.227$ on S_{10}), approaching the closed form as the large-q truncation recedes. We do *not* claim a normal law: at the accessible scale $\ln\ln(6N) \approx 3$ the distribution has not reached its Erdős–Kac limit (mean $\neq$ variance, nonzero skew), and the limiting law is left open. No claim is made about the infinitude of twin primes.

1 The single-prime probability

For an ordinary integer, a prime q divides it with probability $1/q$. A twin centre N is constrained: $6N \mp 1$ must both be prime, so modulo $q > 3$ the residue of N must avoid the two forbidden classes $\text{dead}(q) = \{\pm 6^{-1} \bmod q\}$, leaving $q-2$ admissible residues. The residue 0 (i.e. $q \mid N$) is one of these admissible classes, and within them no class is locally favoured for twin-ness at q. Hence:

Theorem 1 (single-prime divisibility)**.** *For $q > 3$,* $P(q \mid N \mid N \text{ twin}) = \dfrac{1}{q-2}$.

Measured on S_{10} (23,988,173 twin centres) and S_9, the equality holds per prime to high precision (Table 1, Fig. 1 left): for $q \le 401$ the ratio of the measured probability to $1/(q-2)$ is within 0.4% (0.1% for the smallest, most frequent q), tightening from S_9 to S_{10}. For large q the per-prime measurement is dominated by sampling depletion ($P \sim 1/q$ gives only tens of divisible centres at $q \sim 10^5$), and is not a deviation from the law.

2 Root of the Part I enrichment

Theorem 1 is the microscopic source of the macroscopic enrichment of Part I. The relative density of twin centres among integers divisible by q over those not is, from the raised probability,

$$\frac{P(q \mid N \mid \text{twin})}{P(q \nmid N \mid \text{twin})} \Big/ \frac{1/q}{1-1/q} = \frac{q-1}{q-3},$$

q	$1/q$ (ord.)	$1/(q-2)$	meas S_{10}	ratio
5	0.20000	0.33333	0.333349	1.0000
7	0.14286	0.20000	0.200004	1.0000
11	0.09091	0.11111	0.111090	0.9998
13	0.07692	0.09091	0.090895	0.9998
17	0.05882	0.06667	0.066597	0.9990
19	0.05263	0.05882	0.058812	0.9998
23	0.04348	0.04762	0.047595	0.9995
47	0.02128	0.02222	0.022217	0.9998
97	0.01031	0.01053	0.010521	0.9994
401	0.00249	0.00251	0.002517	1.0044

Table 1: $P(q \mid N \mid N \text{ twin})$ on S_{10} equals $1/(q-2)$, not the ordinary $1/q$. Small q confirm the law to $\leq 0.4\%$; large q (not shown past 401) are sampling-limited.

so the conditional density carries the factor $\prod_{q|N}(q-1)/(q-3)$ first measured in Part I. The enrichment is precisely the product of the single-prime probability boosts $1/q \to 1/(q-2)$.

3 The mean shift

Since $\omega_{>3}(N) = \sum_{q>3} \mathbf{1}\{q \mid N\}$, the mean shift between twin centres and ordinary integers is, term by term,

$$\langle\omega_{>3}\rangle_{\text{twin}} - \langle\omega_{>3}\rangle_{\text{ord}} = \sum_{q>3}\Big(\frac{1}{q-2} - \frac{1}{q}\Big) = 0.2604,$$

a convergent constant (each term $\sim 2/q^2$), independent of any cutoff—unlike the individual means, which diverge as $\ln\ln$. The measured finite-N shift is smaller and grows with the shell: $+0.2220$ on S_9, $+0.2270$ on S_{10} (Table 2). The deficit from 0.2604 is the contribution of large primes q whose divisibility is truncated by the finite range $N \leq H$; as H grows these terms are recovered, and the two-shell trend confirms the approach to the closed form.

shell	ord. mean	twin mean	shift	twin var	twin skew	twin ex.kurt
S_9	2.3599	2.5819	+0.2220	0.9423	0.246	−0.326
S_{10}	2.4757	2.7027	+0.2270	1.0394	0.258	−0.287

Table 2: Mean shift grows toward the closed form 0.2604 with shell depth. The normality diagnostics (mean $\neq$ var, nonzero skew/kurtosis) show the Erdős–Kac limit is not reached at this scale.

4 Normality is not claimed

One might expect $\omega_{>3}$ of twin centres to follow an Erdős–Kac normal law with Bernoulli parameter $1/(q-2)$. We do *not* assert this. At the accessible scale the mean (2.70) and variance (1.04) are far apart, and the skewness (+0.26) and excess kurtosis (−0.29) are nonzero; with $\ln\ln(6N) \approx 3$ the distribution is far from its limit, for ordinary integers as much as for twins. The limiting law of $\omega_{>3}$ over twin centres—and whether the natural variance $\sum 1/(q-2)\big(1 - 1/(q-2)\big)$, constrained by the finite number of prime factors, governs it—is left as an open problem.

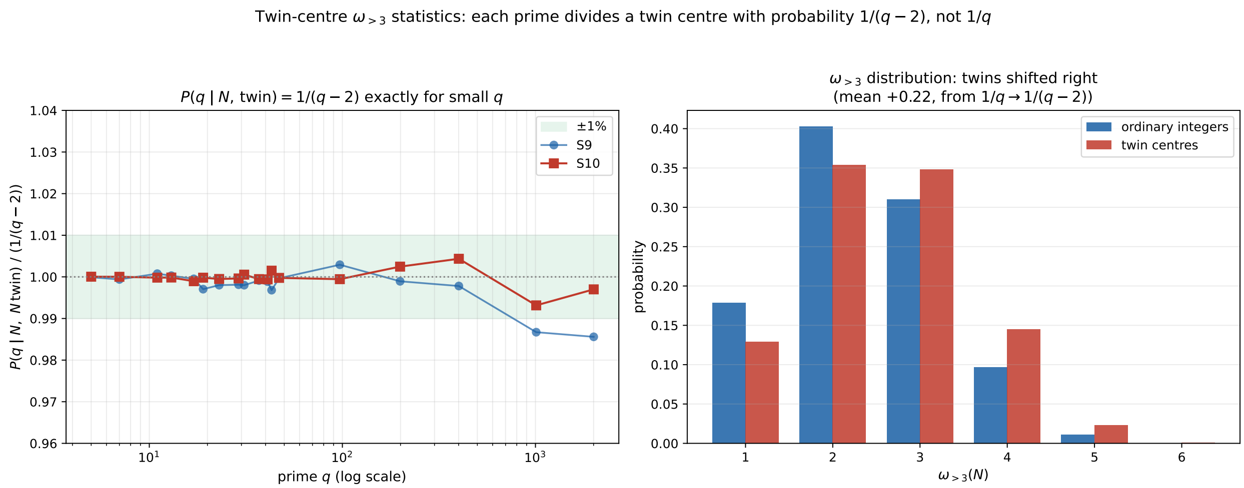

Figure 1: Left: $P(q \mid N, \text{twin})/(1/(q-2))$ versus q, both shells; flat at 1 for small q, sampling-limited at large q. Right: the $\omega_{>3}$ distribution of twin centres (red) is shifted right of ordinary integers (blue).

5 Conclusion

Twin centres carry a clean factor statistic: each prime $q > 3$ divides one with probability $1/(q-2)$ rather than $1/q$, verified per prime on S_9 and S_{10} to within 0.4% for the dominant small primes. This is the microscopic origin of the Part I enrichment $\prod(q-1)/(q-3)$, and it shifts the mean $\omega_{>3}$ rightward by the convergent constant $\sum_{q>3}(1/(q-2)-1/q) = 0.2604$, approached from below as the shell deepens. We make no normality claim: at this scale the Erdős–Kac limit is not reached, and the limiting distribution is open. As throughout this series, no claim is made about the infinitude of twin primes or any k-tuple conjecture; this is a measured, factor-resolved account of the arithmetic of the twin skeleton.

Data and code availability. The per-prime probabilities and the ω distribution are openly available [5]; the S_{10} twin count 23,988,173 matches Part I.

References

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